EL NIÑO ACTION BRIEFING · 26 SEPTEMBER 2026 · VERIFIED RUN · COFFEE-INDUSTRY ACTIONS FIRST

Matagalpa, Nicaragua

The coffee industry's channels reach the part of Matagalpa this El Niño hit least. The highlands had normal rain and face a quality and labour squeeze. The drought is in the Sébaco west, where no coffee contract reaches, and the city drinks from the same shrinking aquifer that irrigates the valley's rice.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Matagalpa department, about 613,000 people in 13 municipalities (about two Climate Impact Lab regions)	No figure published for Nicaragua	Jun 2026 – Feb 2027 (heat only)	West: drought and a falling aquifer. Highlands: quality loss and pickers leaving. East: landslides

14/14 recommendations verified. 7 corrected (partially confirmed), 0 confirmed outright, 2 unverified, 5 contraindicated. The funding target moved from $311k to $298k, down 4%, so the envelope stands. **The run's central disagreement was settled against the coffee channel:** no source shows that households in Matagalpa's drought zone pick coffee, so the claim that estate payrolls carry money into the dry west is unproven and is not made here. **Ranking rule:** coffee-industry actions first, at the reader's request, followed by what reaches the drought zone. **Resolution caveat:** a department, not a town. **Method**

* Suggested formulation: check with a health worker before sharing.

EXECUTIVE SUMMARY — AT A GLANCE



12.93°N 85.92°W, Nicaragua · Natural Earth

- **The risk.** Drought and a falling aquifer in the Sébaco west, whose well field supplies over 92,000 city residents; on the slopes, Iota's November 2020 Peñas Blancas debris flow killed about 10 people.
- **First action.** Importers and roasters spread purchases across at least three exporters or buy direct, pay early, and each commit to a written farm-gate price, in contracts before November.
- **The money.** $98k floor, $298k target, about $667k ceiling; UNICEF under-5 malnutrition treatment in the dry west, $150k, is the highest-value dollar.
- **Do not.** Do not count coffee money as covering the drought: it lands in highlands with near-normal rain, leaving the Sébaco west with nothing while appearing covered.

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caveat: web search was exhausted; agents worked by direct fetch. National facts carry over from the verified Ocotal and Jinotega runs.

EXPOSURE, HONESTLY

THREE MATAGALPAS

- **The coffee highlands** (Matagalpa, San Ramón, El Tuma–La Dalia, Rancho Grande; farms at 800–1,450 m) **had near-normal rain.** El Tuma–La Dalia and Río Blanco are named among the humid zones with normal or above-normal rain (Confidencial, 25 September). The hazard is quality and price: heat, borer, and unfilled beans, which cut Nicaragua's 2023/24 output 10% in the last El Niño.
- **The dry west** (Sébaco, Ciudad Darío, Terrabona, San Isidro, Esquipulas, San Dionisio) sits in the Dry Corridor, where August was the worst month and the shortfall is expected through October. Sébaco is one of Central America's largest rice producers and also grows beans, onion, carrot, and beet. No 2026 loss figures for these municipalities were found; rice growers in Sébaco and San Isidro report pest damage.
- **The city's water is the dry west's water.** The Chagüitillo well field in Sébaco supplies more than 92,000 Matagalpa residents. In June 2026 the water utility's departmental delegate said the Aranjuez, Molino Norte, and Chagüitillo sources "have drastically reduced their flow"; some barrios get water by the hour or at night; six new wells are planned at C$29 million. Studies cited in the same report show water-table declines of up to 9.1 metres a year (2010–18), and irrigation draws 73.5 million cubic metres a year against 6.1 million for drinking water (Mosaico CSI, 16 June 2026).
- **The humid east and the slopes.** Iota's debris flow at San Martín de Peñas Blancas in November 2020 killed about 10 people with 2 missing (reports range from 9 to 12); two days later three siblings, aged 22, 15, and 9, died in a mudflow at Cerro El Puyú between Waslala and Mulukukú.
- **Labour.** Estates now recruit pickers door to door ("we have to go and find them in their homes"); day rates run C$268–320, some producers are cutting area rather than raise pay, and one farm's crew fell from 80 at peak to 11–18. Remittances rose from $1.5 billion in 2018 to $6.2 billion in 2025 (Confidencial, 28 June 2026). **Where Matagalpa's pickers come from is unknown**; estates trucked workers from León, Chinandega, and Managua last season.
- **Where the money goes.** One Matagalpa producer was paid a flat $200 per hundredweight through 2024 while the international price rose from about $235 to $268.6, as exporters' "processing plus taxes" deduction grew from $35 to $50–52. When CISA closed in December 2023, exporters paid C$400 less per hundredweight, and its lender still held growers' mortgage deeds, blocking new credit elsewhere.

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO

COFFEE INDUSTRY FIRST

The first six actions have a coffee-industry owner or payer and protect the highlands and the people who pick there. Actions seven and eight reach the drought zone, which the coffee channel does not. The Historian's finding frames the first group: in Matagalpa, every crisis has landed hardest on people who work coffee they do not own, and the only channel that has reached them is the employer's payroll.

Buyers: spread purchases across at least three exporters or buy direct, pay early, and commit individually to a written farm-gate price

PARTIALLY CONFIRMED

ECONOMIST · SKEPTIC · HISTORIAN · EMERGENCY MGR

ADAPTATION INVENTORY: PRECEDENT, US CONTINGENT LOANS KEPT FIRMS OPEN

OWNER	BY WHEN	COST	PREVENTS
Importers and roasters buying via Atlantic, Olam, and others	Contracts before November	About $15–20/qq moved from exporter margin to grower (estimate)	The widening discount growers pay when buyers are few

* Suggested formulation: check with a health worker before sharing.

The Economist's one thing: the drought is not uninsured, it is paid for as a discount, and after CISA closed that discount widened because there were fewer buyers, not because costs rose. The cheapest insurance in Matagalpa is another buyer bidding at the dry mill. A farm-gate price written into the contract, as each buyer's own commitment, makes the discount visible. **Do not coordinate a share with other buyers**: the Specialty Coffee Transaction Guide publishes price data but explicitly makes no pricing recommendations, for antitrust reasons. No precedent for an 80% farm-gate clause was found.

Estates: pay pickers in full before the Christmas pause, through the payroll

PARTIALLY CONFIRMED | HISTORIAN · FRONTLINE · EMERGENCY MGR · PHILANTHROPIST · ECONOMIST

ADAPTATION INVENTORY: PRECEDENT, NICARAGUA CASH TRANSFER PLUS PRODUCTIVE GRANT

OWNER	BY WHEN	COST	PREVENTS
Mierisch's 8 Matagalpa farms (618 ha); Selva Negra and La Hammonia; importers via contract price	Commit by 15 November	$71k: 390 Mierisch pickers × 100 days × $1.50 + La Hammonia early pay at $0.75	Crews leaving in December, and cherry over-ripening in a year of unfilled beans

$1.50 is 17–21% of a C$268–320 day. La Hammonia already feeds its pickers at harvest, so its share is early pay only. The picker counts use 0.63 per hectare from one farm; no estate publishes picker numbers. This protects the highland harvest and the pickers wherever they come from. It is not evidence of money reaching Matagalpa's own dry west.

Lenders: a deferral clause in 2026/27 pre-harvest credit, backed by a first-loss reserve held outside Nicaragua

PARTIALLY CONFIRMED | ECONOMIST · EMERGENCY MGR · HISTORIAN · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	TRIGGER
Exporters, impact lenders, importers	31 October	$50k reserve; carry on deferred balances ~10% (estimate)	Highlands: quality downgrade. Dry west: delivered volume below 80% of contract

* Suggested formulation: check with a health worker before sharing.

Matters most where a failed lender still holds growers' deeds. With near-normal highland rain the trigger is unlikely to fire, so the reserve is likely to come back; held outside Nicaragua, it avoids the foreign-funding question. The rust study (Villarreyna 2020) covered San Ramón and El Tuma–La Dalia: technical fixes were "not sufficient" where farmers lacked money.

Buyers: price unfilled beans one grade down instead of rejecting them this season

UNVERIFIED · NO PRECEDENT FOUND | ECONOMIST · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Importers and roasters	Before harvest	Grade differential on the affected share, $10–30/qq (estimate)	Losing the income El Niño takes through grading

Graded differentials are standard practice; a season-specific commitment not to reject heat-damaged lots has no found precedent. It is the highland version of the Jinotega recommendation to hold the price tier.

Estates: harvest-camp care: surge supplies for the La Hammonia clinic, water filters, barrel lids, and shade and water for pickers trucked in from the hot lowlands

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Estates, invoiced as a sustainability premium	Before pickers arrive	$12k clinic supplies; $15k filters (price and household count unconfirmed)	Diarrhoea, dengue, and heat strain on already-damaged kidneys

* Suggested formulation: check with a health worker before sharing.

La Hammonia is the strongest delivery point found in any Nicaragua run: about 300 workers live on the farm with their families in concrete houses, a school has run since 1975, and the estate funds a full-time nurse. The nurse is already there, so money buys oral rehydration salts, zinc, vector control, and referral transport. In Managua, uncovered barrels had 73% more mosquito pupae. Kidney disease was near absent in a 1,000-metre Matagalpa coffee village (0% of men), so the risk arrives with lowland pickers, not the highlands. Filters are held until a supplier quote exists; the Nicaraguan ceramic-filter factory's status is unclear.*

Estates on the Peñas Blancas massif and in the humid east: map worker houses below slide-prone slopes and move them uphill before late 2027

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | ADAPTATION INVENTORY: PRECEDENT, NICARAGUA BRIDGES OFFSET FLOOD INCOME LOSS

OWNER	BY WHEN	COST	PREVENTS
Estates on private land	Before the 2027 hurricane season	$500–2,000 per house (estimate)	A repeat of 2020's slope deaths

Plans treated Iota as a coastal storm; the deaths came far inland on steep coffee-forest slopes. Iota and Mitch both struck in La Niña; if La Niña follows this El Niño, late 2027 is the window. Acting on private estate land keeps it politically safe.

The shared aquifer: the water utility and Sébaco's irrigators agree a drinking-water priority for February to April, before the city's wells fail

PARTIALLY CONFIRMED | EMERGENCY MGR · VERIFICATION | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Water utility (ENACAL) and the municipality; Sébaco rice and vegetable irrigators; wet mills in the city	Agreement by January; watch well levels from January	Negotiated; six new wells already planned at C$29 million	92,000 city residents losing supply at the harvest peak

The Emergency Manager's one thing, confirmed and sharpened: Matagalpa's taps depend on the Sébaco aquifer, which irrigation draws about twelve times harder than the city, and whose water table has fallen up to 9 metres a year. The shortage will show in the valley's wells from February to April, when the harvest brings the most people into the city. Ask about Sébaco well levels in January, not city

* Suggested formulation: check with a health worker before sharing.

rainfall. Coffee's part is small but real: wet mills in and around the city can cut their draw.

For the dry west's children: fund malnutrition treatment through UNICEF, released only once UNICEF confirms it can target those municipalities

PARTIALLY CONFIRMED · CONDITIONAL | PHILANTHROPIST · PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
UNICEF with the health ministry	For the November–March peak	$150k	Under-5 wasting from the income shock in the drought zone

The Public Health lens's one thing: the deaths are unlikely to come from highland heat; they will come from income loss in the Dry Corridor showing up as child wasting by January. Treating severe acute malnutrition costs $26–53 per healthy life-year. This is the highest-value dollar in the report and it is not a coffee dollar. WFP school meals probably cover school-age children in these municipalities; the gap is under-5s. Nothing published confirms that UNICEF can earmark funds to Matagalpa's dry municipalities under its government agreement.

Growers and buyers: press for the coffee export levy's renovation fund to be used before the 2027 rains

UNVERIFIED | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Growers' associations	2027	Existing fund	An old, rust-prone tree stock facing the next shock

Law 853 (2013) levies about $4 per 60-kg bag for a renovation fund; industry sources estimate more than $40 million collected since 2013, and growers have "mixed opinions" (USDA). How the fund is controlled and spent could not be confirmed.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Importers & roasters	Exporters (Atlantic, Olam)	Impact lenders	Estates (Mierisch, Selva Negra, La Hammonia)	La Hammonia clinic	Growers' associations	Funders	ENACAL (water utility)	Municipality	Sébaco irrigators	Wet mills	UNICEF	Health ministry
1. Spread buying across exporters, written farm-gate price	**A/R**	**C**		**I**									
2. Pay pickers in full before Christmas	**R**			**A/R**			**C**						
3. Deferral clause backed by offshore first-loss reserve	**R**	**A/R**	**R**	**I**			**C**						
4. Price unfilled beans one grade down	**A/R**	**I**		**I**									
5. Camp care: La Hammonia clinic, filters, shade	**R**			**A/R**	**R**		**C**						
6. Move worker houses off slide-prone slopes				**A/R**									
7. Drinking-water priority on the Sébaco aquifer								**A**	**R**	**R**	**R**		
8. Dry-west child malnutrition treatment via UNICEF							**R**					**A/R**	**R**
9. Use the export levy's renovation fund	**R**					**A/R**							

What the chart shows. **Estates are accountable for three of the nine rows, through three named estate groups.** Rows 2, 5, and 6 run through Mierisch, Selva Negra, or La Hammonia payrolls and land, and the report warns against routing the season through one estate group. **None of the six coffee rows reaches the drought zone.** Only rows 7 and 8 do, and they are accountable to a water utility and to UNICEF, not to coffee firms. **Row 8 depends on a confirmation nobody has published.** It is the highest-value dollar in the report, but nothing shows that UNICEF can earmark funds to the dry municipalities. **Row 9 has an owner but no visible controller.** Growers' associations are named, but how the levy fund is controlled and spent could not be confirmed.

WHERE MONEY DOES THE MOST

FUNDING

Verified and corrected. The coffee channel is funded as the politically safe route for the highlands and pickers; the highest-return dollar, for the dry west's children, goes through UNICEF, and the coffee channel is a complement to it, not a substitute.

FLOOR

$98k

TARGET

$298k

CEILING

About $667k

LARGEST UNCERTAINTY

Whether UNICEF can take $150k earmarked for the dry municipalities

* Suggested formulation: check with a health worker before sharing.

Arithmetic. Floor: La Hammonia clinic supplies $12,000 + filters $15,000 + picker early pay $71,250 (Mierisch 390 × 100 × $1.50 = $58,500; La Hammonia 170 × 100 × $0.75 = $12,750) = $98,250. Target: floor + UNICEF $150,000 + first-loss reserve $50,000 = $298,250. Ceiling: extended picker support across about 3,000 hectares of estates, UNICEF to $250,000, reserve to $100,000, filters and clinics to $45,000, less the duplicated meals, about $667,000. The target fell 4% from the lens's $311k.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
UNICEF with the health ministry	Action 8: under-5 malnutrition treatment in the dry west	$150k	Earmarked contribution, released on written confirmation	$26–53 per healthy life-year; the only line reaching the drought zone	Keep, conditional
Mierisch estates; La Hammonia	Action 2: picker early pay	$71.3k	Contract-price differential through payroll	17–21% of a day's pay; La Hammonia already feeds pickers	Keep, reduced
Buyer or exporter, held outside Nicaragua	Action 3: first-loss reserve	$50k	Reserve behind a deferral clause	Likely returned: highland rain near normal	Keep
La Hammonia clinic	Action 5: surge supplies, vector control, referral transport	$12k	Invoiced sustainability premium	Nurse already estate-funded; money buys supplies	Keep, reframed
Selva Negra and Mierisch households	Action 5: ceramic filters	$15k	Estate operating cost	Diarrhoea RR 0.39; price and household count unconfirmed	Hold for quote

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Do not count coffee money as covering Matagalpa's drought

CONTRAINDICATED SKEPTIC; CONFIRMED IN VERIFICATION

The highlands where coffee money lands had near-normal rain. The drought is in the Sébaco west, which grows rice, vegetables, and cattle. No source shows its households pick coffee; estates recruited from León, Chinandega, and Managua. Claiming coffee payrolls as a "transfer pipe" into the drought zone would leave the worst-hit area with nothing while appearing covered.

Instead: judge coffee money by what it does for quality and pickers in the highlands; fund the dry west through actions 7 and 8.

Do not route anything through the diocese, parishes, or Church radio

CONTRAINDICATED SKEPTIC · FRONTLINE; CONFIRMED

On 1–2 August 2022 the telecoms regulator closed ten diocesan radio stations, including Radio Católica de Sébaco and Radio Católica de Waslala; police used tear gas on residents defending the Sébaco priest. Bishop Rolando Álvarez was sentenced to 26 years and expelled. The diocese still has 28 parishes and about 615,700 baptised Catholics, but anything named puts the people on it at risk. The Frontline lens: anyone who keeps a list of names is a risk to everyone on it; never write the parish's name on anything.

* Suggested formulation: check with a health worker before sharing.

Instead: the foreman's truck route and the catechist's chapel rounds, carrying messages, not lists.

Do not route the season through one exporter or one estate group

`CONTRAINDICATED` `SKEPTIC · ECONOMIST; CONFIRMED`

CISA handled about 1.5 million hundredweight a year; its closure cut prices C$400 per hundredweight and left growers' deeds with its lender. Matagalpa's named channels are few: Mierisch, Selva Negra, and CECOCAFEN, whose websites no longer resolve and whose current status could not be confirmed.

Do not replant on credit, and do not use a department-wide drought trigger

`CONTRAINDICATED` `SKEPTIC; CARRIED FROM JINOTEGA`

Young plants are the most drought-vulnerable and replanting means three years of debt. A department-wide rainfall average would pay the humid highlands for a drought they did not have and miss the west. Use quality triggers in the highlands and rainfall or volume triggers only in the west.

Do not coordinate a farm-gate price share with other buyers

`CONTRAINDICATED AS FRAMED` `VERIFICATION`

A shared percentage across buyers raises antitrust exposure, which is why the industry's own price-transparency guide makes no pricing recommendations. Each buyer commits for itself.

* Suggested formulation: check with a health worker before sharing.

WHAT HAS HAPPENED HERE BEFORE

THE RECORD

DATE	EVENT	TOLL	WHAT FOLLOWED
19th century	Ludwig Elster and Katharina Braun plant the first coffee; more than 120 European immigrants follow	—	Estates on Indigenous land
Mar and Aug 1881	Indigenous revolt over forced cabildo, road, and telegraph labour; city besieged; two telegraph operators hanged	"Harshly repressed, at the cost of many deaths"	Coffee economy consolidated
2001	Coffee price collapse	Pickers laid off first (road camps and deaths unverified)	Red de Protección Social in Matagalpa protected household spending
17 Oct 2007	Worst recorded flood of the Río Grande de Matagalpa through city barrios	Deaths reported	—
2012–14	Coffee rust across Central America	Yields −10–55%; ~$515M losses; food insecurity and migration	Only ~14% of Nicaraguan arabica replanted since
Nov 2020	Iota: Peñas Blancas debris flow; Cerro El Puyú mudflow	~10 dead, 2 missing; 3 siblings	—
Aug 2022	Ten diocesan radios closed; Sébaco parish raided	—	Bishop Álvarez imprisoned, expelled 2024
Dec 2023	CISA stops buying before harvest	Prices down C$400/qq; deeds held by its lender	Fewer buyers, wider deductions
2024	Exporters' deduction rises from $35 to $50–52/qq as world price climbs	—	—
Jun 2026	Water utility admits city sources "drastically reduced"	Barrios on hourly or night supply	Six new wells planned

* Suggested formulation: check with a health worker before sharing.

DATE	EVENT	TOLL	WHAT FOLLOWED
Aug–Sep 2026	Dry Corridor deficit; highland rain near normal	—	This briefing

The recurrence nobody has noticed, from the Historian: every crisis in Matagalpa has landed hardest on people who work coffee they do not own, from the conscripted road builders of 1881 to the laid-off pickers of 2001. Relief has always been routed to owners. The one channel that has reached labourers is the employer's payroll, and the record shows no second one.

WHERE THE EXPERTS DISAGREE

RESOLVED AND OPEN

- **Coffee payrolls as a transfer pipe to the dry west** versus **coffee channels miss the drought**. Settled toward the Skeptic: no evidence links the dry west's households to coffee picking.
- **Selva Negra publishes no worker data** versus **it does**. Settled: the data are on its history and La Hammonia pages, not its homepage.
- **Open:** whether UNICEF can target Matagalpa's dry municipalities, and whether WFP's school meals already cover them.

WHAT NOBODY COVERED

BLIND SPOT

No lens found who actually lives from the Sébaco valley's farms: the rice and onion labourers and the small cattle keepers of the dry west are the people most exposed and the least visible in any source reached. No lens found where Matagalpa's pickers come from, which decides whether any coffee money reaches drought-hit homes. No lens looked at the North Caribbean border around Waslala and Mulukukú, where the 2020 slide deaths and the malaria zone meet. And no dengue figures for the department could be found for 2026.

HOW TO READ THIS

METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

* Suggested formulation: check with a health worker before sharing.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

This run. Coffee-industry actions were ranked first at the reader's request, but verification found that the coffee channel does not reach this department's drought zone, and the report says so rather than stretch it. National and Jinotega facts were carried in so lenses could spend effort on local evidence. Web search was exhausted; all twelve agents worked by direct fetch. Five verifiers checked fourteen recommendations.

The El Niño projection covers *heat only* and publishes nothing for Central America. The Adaptation Inventory holds two Nicaragua entries, both cited, and nothing on coffee, aquifers, or harvest labour.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The location was taken as given: Matagalpa, Nicaragua. Climate Impact Lab impact region, as the report records it: Matagalpa department, about 613,000 people in 13 municipalities (about two Climate Impact Lab regions). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (oral rehydration salts, zinc, safe water, and mosquito control in harvest camps) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.

* Suggested formulation: check with a health worker before sharing.

- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (verification status in brackets) · Rainfall 25 Sep 2026 confidencial.digital [confirmed; no Matagalpa municipality named for cattle] · Sébaco rice and pests laprensani.com · crop pests laprensani.com · Chagüitillo and the city's water mosaicocsi.com [found] · water complaints confidencial.digital [Matagalpa named only among complaints] · Matagalpa es.wikipedia · Sébaco es.wikipedia · Río Grande de Matagalpa es.wikipedia · Pickers door to door confidencial.digital · harvest pause, 28 Dec 2025 confidencial.digital [confirmed with corrections] · Exporter deductions, 27 Sep 2024 confidencial.digital · CISA suspension, 6 Dec 2023 confidencial.digital · USDA GAIN NU2025-0004 · Specialty Coffee Transaction Guide transactionguide.coffee [confirmed] · Mierisch farms fincasmierisch.com · La Hammonia workers lahammonia.com · Selva Negra history selvanegra.com [confirmed] · WFP Nicaragua wfp.org · UNICEF Nicaragua unicef.org · Harvey et al. 2021 PMC8406019 · Ewbank et al. 2019 doi.org · Villarreyna et al. 2020 doi.org [confirmed] · 1881 revolt es.wikipedia · first coffee wikipedia · diocesan radios closed confidencial.digital · Diocese of Matagalpa es.wikipedia [confirmed] · Iota slides confidencial.digital [toll 9–12 across reports] · Kidney disease in a coffee village PMC4405255 · young workers' kidney function PMC13483704 [eGFR<90, not <60] · Managua barrels PMC13148774 · Salud Mesoamérica PMC8638427 · PMC8720941 [confirmed] · National and Jinotega facts carried from the verified briefings in this folder. Struck or unverified: coffee payrolls as a transfer to the dry west; Mitch flooding in Matagalpa city; "230 mm in 30 hours" (a grower's remark in inches); 2001–02 road camps and deaths; Matagalpa cattle distress-sale figures; pickers' origins; Matagalpa dengue 2026; CECOCAFEN's status; the export levy fund's control.
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions; Nicaragua cash transfer and productive grant entry; Nicaragua bridges entry; US contingent loans entry)
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